4AL3 Project Proposal - Group 28

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1. Team Members

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2. Task Title

Book Review Spoiler Detection Model

3. Task Definition

- Supervised binary text classification
- Unstructured natural language text
- 2 classes, either spoiler or non-spoiler
- Single label, each x is assigned exactly one y (either 1 or 0)

4. The Problem

When trying to find books to read, many people browse reviews before starting a book. There are popular platforms like Goodreads to facilitate this, but reviews often contain major spoilers such as twists, character deaths, or the ending. This can ruin the experience of reading the book and upset readers that just want to hear a high-level overview or opinions. Goodreads lets reviewers hide this content behind spoiler tags [1], but it is based on trust, since humans decide whether to include the spoiler tags or not. Because of this, readers can have a book spoiled before they ever open it even if it doesn't have a spoiler tag.

Real-world impact. Creating a spoiler detector keeps readers safe by checking for untagged spoilers before they are posted, or let readers filter for review that the platform thinks contains spoilers. This would make review platforms more useful and "safe" to readers who have not yet read the book. If our model is well-trained to be generalizable, we can extend the model beyond books to detect spoilers on reviews and discussions of movies, TV shows, and video games.

Why it is challenging. Spoiler detection is a unique case of text classification that poses a few challenges

- *Context dependencies* Whether something is a spoiler or not is heavily dependent on the context of the book. Something that might be a non-spoiler premise about the book may be flagged about the

spoiler just because the model thinks it's important information.

- *Book-specific vocabulary.* Wan et al. found that places, people, and proper names are strongly linked to spoilers [2]. A model trained with these names can fail on books with names it hasn't seen, because it associates those names with spoilers.
- *Class imbalance.* Only a small percentage of the data contains spoilers, so the baseline model that always predicts "no spoiler" would reach high accuracy even if it doesn't find any spoilers.
- *Noisy labels.* Labels come from users either flagging the review as spoiler or non-spoiler. Because of this, there might be reviews that do have spoilers, but the user forgot to tag them as such.

5. Data Sources and Plan for Data Collection

The dataset we are planning on using is the Goodreads Spoilers Dataset, originally collected and released by McAuley Lab at UC San Diego, and reposted on Kaggle by Ahmad Mujtaba.

- Original source: `https://cseweb.ucsd.edu/~jmcauley/datasets/goodreads.html`
- Kaggle link: `https://www.kaggle.com/datasets/pypiahmad/goodreads-book-reviews`

The data was collected in late 2017 from goodreads.com by scraping users' public Goodreads shelves, i.e. pages visible without login, for academic use. The original data consists of three groups of datasets: meta-data of the books, user-book interactions (users' public shelves), and users' detailed book reviews. For our project, we will only use the third.

Metadata includes reviews (the text of the reviews provided by users), ratings (assigned to the books by users), and spoilers (annotated spoilers within the review text). The spoiler labels ultimately originate from Goodreads' own spoiler tag feature [1] [2], and we determine whether the entire review is a spoiler or not based on the spoiler tag.

Given the dataset's size of 1,378,033 reviews (of 25,475 books by 18,892 users), we plan to use a smaller, randomly sampled subset of the data because of pure computational limits.

6. Expected Size of Dataset

- Expected size of dataset: 10,000
- Training data: 8,000 (80%)
- Validation data: 1,000 (10%)
- Testing data: 1,000 (10%)

The original dataset has ~1.3 million datapoints; we will parse through it and only use 10,000 datapoints

instead.

Below are 3 datapoints from the dataset. The biggest thing to note is that spoilers will contain `(view spoiler)` and `(hide spoiler)` in the text, and that will be used to identify spoilers.


```
{'user_id': '8842281e1d1347389f2ab93d60773d4d', 'book_id': '18245960', 'review_id': 'dfdbb7b0eb5a7e4c26d59a937e2e5feb', 'rating': 5, 'review_text': 'This is
      a special book. It started slow for about the first third, then in the middle third it started to get interesting, then the last third blew my mind.
      This is what I love about good science fiction - it pushes your thinking about where things can go. \n It is a 2015 Hugo winner, and translated from
      its original Chinese, which made it interesting in just a different way from most things I\'ve read. For instance the intermixing of Chinese
      revolutionary history - how they kept accusing people of being "reactionaries", etc. \n It is a book about science, and aliens. The science described
      in the book is impressive - its a book grounded in physics and pretty accurate as far as I could tell. (view spoiler)[Though when it got to folding
      protons into 8 dimensions I think he was just making stuff up - interesting to think about though. \n But what would happen if our SETI stations
      received a message - if we found someone was out there - and the person monitoring and answering the signal on our side was disillusioned? That part
      of the book was a bit dark - I would like to think human reaction to discovering alien civilization that is hostile would be more like Enders Game
      where we would band together. \n I did like how the book unveiled the Trisolaran culture through the game. It was a smart way to build empathy with
      them and also understand what they\'ve gone through across so many centuries. And who know a 3 body problem was an unsolvable math problem? But I
      still don\'t get who made the game - maybe that will come in the next book. \n I loved this quote: \n "In the long history of scientific progress, how
      many protons have been smashed apart in accelerators by physicists? How many neutrons and electrons? Probably no fewer than a hundred million. Every
      collision was probably the end of the civilizations and intelligences in a microcosmos. In fact, even in nature, the destruction of universes must be
      happening at every second--for example, through the decay of neutrons. Also, a high-energy cosmic ray entering the atmosphere may destroy thousands of
      such miniature universes...." \n (hide spoiler)]', 'date_added': 'Sun Jul 30 07:44:10 -0700 2017', 'date_updated': 'Wed Aug 30 00:00:26 -0700 2017',
      'read_at': 'Sat Aug 26 12:05:52 -0700 2017', 'started_at': 'Tue Aug 15 13:23:18 -0700 2017', 'n_votes': 28, 'n_comments': 1}
```

```
{'user_id': '8842281e1d1347389f2ab93d60773d4d', 'book_id': '16981', 'review_id': 'a5d2c3628987712d0e05c4f90798eb67', 'rating': 3, 'review_text': 'Recommended
      by Don Katz. Avail for free in December: http://www.audible.com/mt/ellison2?so...', 'date_added': 'Mon Dec 05 10:46:44 -0800 2016', 'date_updated':
      'Wed Mar 22 11:37:04 -0700 2017', 'read_at': '', 'started_at': '', 'n_votes': 1, 'n_comments': 0}
```

```
{'user_id': '8842281e1d1347389f2ab93d60773d4d', 'book_id': '28684704', 'review_id': '2ede853b14dc4583f96cf5d120af636f', 'rating': 3, 'review_text': 'A fun,
      fast paced science fiction thriller. I read it in 2 nights and couldn\'t put it down. The book is about the quantum theory of many worlds which states
      that all decisions we make throughout our lives basically create branches, and that each possible path through the decision tree can be thought of as
      a parallel world. And in this book, someone invents a way to switch between these worlds. This was nicely alluded to/foreshadowed in this quote: \n "I
      think about all the choices we\'ve made that created this moment. Us sitting here together at this beautiful table. Then I think of all the possible
      events that could have stopped this moment from ever happening, and it all feels, I don\'t know..." "What?" "So fragile." Now he becomes thoughtful
      for a moment. He says finally, "It\'s terrifying when you consider that every thought we have, every choice we could possibly make, branches into a
      new world." \n (view spoiler)[This book can\'t be discussed without spoilers. It is a book about choice and regret. Ever regret not chasing the girl
      of your dreams so you can focus on your career? Well Jason2 made that choice and then did regret it. Clearly the author is trying to tell us to
      optimize for happiness - to be that second rate physics teacher at a community college if it means you can have a happy life. I\'m being snarky
      because while there is certainly something to that, you also have to have meaning in your life that comes from within. I thought the book was a little
      shallow on this dimension. In fact, all the characters were fairly shallow. Daniela was the perfect wife. Ryan the perfect antithesis of Jason. Amanda
      the perfect loyal traveling companion, etc. This, plus the fact that the book was weak on the science are what led me to take a few stars off - but
      I\'d still read it again if I could go back in time - was a very fun and engaging read. \n If you want to really minimize regret, you have to live
      your life to avoid it in the first place. Regret can\'t be hacked, which is kind of the point of the book. My favorite book about regret is Remains of
      the Day. I do really like the visualization of the decision tree though - that is a powerful concept. \n "Every moment, every breath, contains a
      choice. But life is imperfect. We make the wrong choices. So we end up living in a state of perpetual regret, and is there anything worse? I built
      something that could actually eradicate regret. Let you find worlds where you made the right choice." Daniela says, "Life doesn\'t work that way. You
      live with your choices and learn. You don\'t cheat the system." \n (hide spoiler)]', 'date_added': 'Tue Nov 15 11:29:22 -0800 2016', 'date_updated':
      'Mon Mar 20 23:40:27 -0700 2017', 'read_at': 'Sat Mar 18 23:22:42 -0700 2017', 'started_at': 'Fri Mar 17 23:45:40 -0700 2017', 'n_votes': 22,
      'n_comments': 0}
```


7. Proposed Solution

7.1 Approach

Since the problem is classifying texts into spoilers and non-spoilers, this is a binary text classification problem, i.e. there are just two options for classification.

The general approach is a 4 step plan:

1. Label the reviews as either spoiler or non-spoiler using the metadata / spoiler tags in our data set, and then remove the spoiler tags from the text.
2. Convert the text into numerical features to use as input to our model (like bag-of-words).
3. Start with logistic regression and test on new books in a different test set.
4. Once we have a strong evaluation pipeline, mask words that are specific to the book; since we want the model to be general enough to apply to any book, we should mask character names and places

so that the model does not train on the names but rather learns what a spoiler looks like. If we have time, we can do a neural network model as well.

We were inspired by Wan et al. for the masking step of the solution, because their research found that book-specific words were very indicative of spoilers [2]. Training a model without masking the book-specific words could have the model learn rules like "reviews that mention Dumbledore are a spoiler", and thus be less generalizable to new test sets.

One way we identified to pre-process the data is by using NER (Named Entity Recognition) – there are public libraries that are able to replace proper names such as places and people with `[LOCATION]` and `[PERSON]` tags.

We will also investigate masking specific words that only show up in a single book, i.e. made-up city names are likely to just show up 5 times throughout all the data. We could thus create a Python script that checks for words that appear in at least 5 reviews, but a large majority (90%) of those reviews come from the same book; then we can assume that it is a book-specific term and mask it with `[TERM]`.

7.2 Features and Target Labels

Since this is a binary text classification problem, the target label is going to be 1 if the review contains a spoiler and 0 if it does not contain a spoiler.

The features would be from the review text (as mentioned in the approach).

We will use a bag-of-words representation weighted by TF-IDF (term frequency–inverse document frequency) [6].

This makes it so that a word gets a higher score if it is frequent in a review (the term-frequency part) and if it is rare throughout the set, i.e. only found in a few reviews (the inverse document frequency aspect).

7.3 Models

We have 3 models to implement based on what we will be taught in the course.

Majority-class baseline. This model is a "dumb model" and will always predict 0, no spoiler. Thus, this is the baseline performance that the rest of our models have to beat.

Logistic regression. Running logistic regression on TF-IDF features is a common text classification technique. It is also easy to inspect the word weights. We can compare different training sets, i.e. training a model on unmasked data and also training a model on masked data.

A neural network. At this point in the course, we have not covered neural networks, but referencing the course outline, W4 and W8 cover neural networks and convolutional neural networks. As such, we would like to implement a neural network to approach this classification problem to apply our knowledge from the course.

7.4 Existing Solutions

Wan et al. [2] built a neural network called SpoilerNet that labels individual sentences in reviews as spoilers.

Boyd-Graber et al. [5] showed that standard machine learning classifiers using word features can detect spoilers in social media posts about TV shows and movies.

Our project is a simpler version of this task for just book reviews, and we want the model to be generalizable to books that the model has not seen.

7.5 Sources of Inspiration

Our solution draws on four sources:

- Wan et al. [2] provided the dataset. Their findings about book-specific words directly inspired our masking step.
- Pang et al. [3] showed that bag-of-words features with standard classifiers work well for binary classification of reviews, which motivates our logistic regression baseline.
- Joachims [4] showed that text classification benefits from using many word features at once, which supports our use of TF-IDF.
- Boyd-Graber et al. [5] showed that spoiler detection is achievable with standard machine learning classifiers.

7.6 Evaluation

We are going to use three sets, a training set, a validation set, and a test set, so we can test the model on books that the model has never seen. The data will be split by book, so that all reviews for a given book only appear in one of the three sets.

We don't want to evaluate the model just on accuracy alone, because most of the reviews are non-spoiler, so even the majority-class baseline model will score 90% accuracy on a set where 90% are non-spoiler.

We will use the F1 score on the spoiler class, which is a score that balances precision and recall. Precision is a measure of how many of the flagged spoilers are actually spoilers, i.e. how many false alarms. Recall is a measure of how many spoilers the model actually catches out of all the spoilers.

A model can be counted as good if it can beat the majority-class baseline on the F1 for the unseen book test set. We can also interpret the weight words in the logistic regression models to confirm the language that the model relies on to detect spoilers. We can read through the mistakes, i.e. misclassified reviews, to evaluate instances where the model is likely to fail.

7.7 Libraries

We will implement the project in Python using:

- pandas and NumPy for data handling
- spaCy for named entity recognition
- scikit-learn for TF-IDF, the majority-class and logistic regression models, the book-based split (`GroupShuffleSplit`), and evaluation metrics
- PyTorch for the neural network
- Matplotlib for plots

8. TA Feedback

We consulted our TA, Abdelaziz Serour, about our project proposal. He agreed that the project is appropriate for the course. He pointed out that our proposal referred both to individual sentences and to full reviews as datapoints. Based on this, we have decided that each datapoint will be a full review, labelled as a spoiler if it contains any spoiler-tagged text. He stressed that the Goodreads spoiler markers must be removed from the input after they are used to create the labels, so the model cannot learn directly from them. We had planned this, and will now explicitly verify that no markers remain in the processed text. He also advised us to check the spoiler/non-spoiler distribution before committing to a sample size of exactly 10,000 reviews. We will therefore examine the class balance in the full dataset first, and adjust our sample size or sample additional spoiler reviews if there are too few positive examples to train on. Finally, he recommended starting with a simple TF-IDF and logistic regression model and building a strong evaluation pipeline before adding NER masking or a neural network. We have reordered our plan accordingly: we will first build and evaluate the logistic regression baseline, and only then add masking and the neural network. Following his advice, we will evaluate all models using precision, recall, F1, and confusion matrices rather than accuracy alone.

References

[1] Goodreads, "Spoiler Tags," `https://www.goodreads.com/blog/show/261-spoiler-tags`.

[2] M. Wan, R. Misra, N. Nakashole, and J. McAuley, "Fine-Grained Spoiler Detection from Large-Scale Review Corpora," in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, A. Korhonen, D. Traum, and L. Màrquez, Eds., Florence, Italy: Association for Computational Linguistics, Jul. 2019, pp. 2605–2610. doi: 10.18653/v1/P19-1248.

[3] B. Pang, L. Lee, and S. Vaithyanathan, "Thumbs up? Sentiment Classification using Machine Learning Techniques," in *Proceedings of the 2002 Conference on Empirical Methods in Natural Language*

Processing (EMNLP), 2002, pp. 79–86. doi: 10.3115/1118693.1118704.

[4] T. Joachims, "Text Categorization with Support Vector Machines: Learning with Many Relevant Features," in *Machine Learning: ECML-98*, Lecture Notes in Computer Science, vol. 1398, Springer, 1998, pp. 137–142. doi: 10.1007/BFb0026683.

[5] J. Boyd-Graber, K. Glasgow, and J. S. Zajac, "Spoiler Alert: Machine Learning Approaches to Detect Social Media Posts with Revelatory Information," in *Proceedings of the American Society for Information Science and Technology*, vol. 50, no. 1, 2013.

[6] G. Salton and C. Buckley, "Term-weighting approaches in automatic text retrieval," *Information Processing & Management*, vol. 24, no. 5, pp. 513–523, 1988. doi: 10.1016/0306-4573(88)90021-0.